2026 3rd International Conference on Image Processing, Multimedia Technology and Machine Learning (IPMML) | 979-8-3195-0802-7/26/$31.00 ©2026 IEEE | DOI: 10.1109/IPMML71439.2026.11682590

BudgetSRE: Budget-Aware Multi-Agent Incident Response for Backend Operations

Qingfan Yang†
Cornell University
Ithaca, USA
angus313074221@outlook.com

Guoze Feng†
Syracuse University
Syracuse, USA
tonyfeng0830@gmail.com

Yihang Wang
University of Southern California
Los Angeles, USA
yihang.wang.dev@gmail.com

Jingyi Chen
Northeastern University
Silicon Valley, USA
joeych.me@gmail.com

Zhiwei Luo*
University of Georgia
Athens, USA
eniacsimon@gmail.com

***Abstract*—Modern cloud incident response increasingly relies on large language model (LLM) powered agents, yet operational deployments face a critical and often overlooked constraint: *cost*. Unconstrained agents can consume thousands of tokens and dozens of tool calls per incident, incurring non-trivial monetary and latency overhead while occasionally issuing unsafe remediation actions. We present BudgetSRE, a multi-agent framework that treats budget—tokens, tool calls, and action depth—as a first-class design parameter alongside response quality and safety. BudgetSRE comprises a lightweight Scout Agent for cheap triage, a Specialist Agent invoked only when diagnostic uncertainty is high, an Action Planner that orders remediations by risk-to-benefit ratio, and a Budget Controller that dynamically reallocates quotas across agents. We formulate budget allocation as a constrained optimization problem and derive a greedy Pareto-style policy. Experiments on the AIOpsLab benchmark show that BudgetSRE achieves a success rate within 4–6 percentage points of an unconstrained oracle at medium budget, outperforms single-agent ReAct by up to 17.9 points at low budget, and reduces unsafe-action incidence by up to 53% compared with naive multi-agent baselines.**

Keywords: AIOps, incident response, multi-agent systems, large language models, root cause analysis, budget-constrained inference

I. Introduction

The operational complexity of modern microservice deployments has spurred considerable interest in autonomous incident response. Frameworks such as AIOpsLab [1] and the broader AgentOps vision [2] articulate a future where LLM-powered agents diagnose faults, identify root causes, and execute remediations without human intervention. Recent systems including RCACopilot [3], FLASH [4], and mABC [5] demonstrate that agents equipped with reasoning and tool-use capabilities [6] can meaningfully outperform rule-based approaches on tasks such as log-metric correlation and service-dependency traversal.

Despite these advances, a critical engineering concern has received insufficient attention: *operational cost*. Frontier LLM APIs charge per token, tool invocations add both monetary and latency overhead, and deep action chains carry non-trivial risk of collateral damage. In practice, SRE teams operate under implicit budgets: a 5-minute mean-time-to-respond window, a per-incident API cost cap, and a risk ceiling that limits how many destructive actions may be attempted before human escalation. Existing benchmarks measure success rate and step count, but few methods explicitly optimize the three-way trade-off among *quality*, *cost*, and *safety*.

We address this gap with **BudgetSRE**, a multi-agent incident responder designed around an explicit budget model. The core insight is that not all incidents are equally hard: straightforward failures should be resolved quickly by a cheap agent, while genuinely ambiguous cases warrant deeper, costlier investigation. BudgetSRE realizes this through four coordinated components: (1) a *Scout Agent* for low-cost initial triage, (2) a *Specialist Agent* invoked only when diagnostic uncertainty exceeds a threshold, (3) an *Action Planner* that ranks remediations by risk-weighted expected improvement, and (4) a *Budget Controller* that dynamically reallocates token and tool-call budgets across agents, enforcing a hard budget ceiling while maximizing expected task success.

We evaluate BudgetSRE on the publicly available AIOpsLab benchmark, covering five fault categories injected into a Kubernetes-hosted social-network microservice. Compared with single-agent ReAct, naive multi-agent orchestration, and fixed-budget truncation, BudgetSRE achieves superior anytime performance curves, lower unsafe-action incidence, and a favorable Pareto frontier in cost-normalized score space. The rest of this paper is organized as follows. Section II surveys related work. Section III presents the framework and budget allocation formulation. Section IV describes experiments and results. Section V concludes.

II. Related Work

LLM Agents for Cloud Operations. Chain-of-thought prompting [7] and the ReAct paradigm enable agents to interleave reasoning and tool use, a pattern adopted by several AIOps systems. Triangle [8] focuses on incident triage with specialized agents, STRATUS [9] proposes a multi-agent architecture for autonomous reliability engineering, and TAMO [10] addresses tool-assisted root cause analysis (RCA) under multi-modal observation data.

† These authors contributed equally to this work.
* Corresponding author

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.

Root Cause Analysis Methods. Earlier RCA work in microservice systems relies on statistical methods, graph neural networks, and anomaly detection [11] [12]. LLM-based approaches have improved reasoning over unstructured logs and multi-modal telemetry correlation [13]. The SWE-agent paradigm [14] demonstrates that effective agent-computer interfaces are crucial for complex multi-step resolution tasks.

Budget-Aware LLM Inference. TALE [15] introduces a two-stage training pipeline to internalize token-budget constraints. BudgetThinker [16] inserts control tokens periodically during inference to track remaining budget. These methods focus on single-turn reasoning compression; BudgetSRE extends the concept to a multi-agent, multi-step incident response setting where budget must be allocated *across agents* rather than within a single generation.

Telemetry-Centric Diagnosis and Drift Handling. Operational evidence pipelines increasingly combine semantic repair, multimodal correlation, and sequence-aware anomaly detection. DriftMender uses LLM-guided semantic repair to maintain real-time ETL pipelines under schema drift [17], while a retrieval-augmented log-metric-trace twin supports evidence-grounded backend failure diagnosis [18]. Protocol anomaly detection from status-code sequences provides a compact behavioral signal for service interactions [19], and self-supervised learning under imbalanced industrial telemetry strengthens fault representations when labeled failures are scarce [20]. Together, these approaches motivate BudgetSRE's selective acquisition of heterogeneous evidence before costly specialist reasoning.

Resource-Aware Agent Reasoning and Scheduling. Problem-centric LLM modeling organizes complex operational decisions around explicit goals, constraints, and alternatives [21]. Resource-efficient fine-tuning through subspace compression and sparse updates reduces the adaptation cost of specialized models [22], while constraint-consistent skill composition improves the reliability of zero-shot agent behavior [23]. At the serving layer, token-level GPU pooling [24] and SLO- and cost-aware elastic scheduling [25] expose resource allocation as an online control problem. Data-centric sequential modeling further emphasizes the value of stateful evidence ordering for session-level decisions [26]. BudgetSRE builds on these principles by coordinating inference, tools, and action depth through a shared incident-level budget.

Calibrated Representation and Risk Scoring. Reliable automated decisions require representations and scores that remain interpretable under uncertainty. AI-assisted expression analysis demonstrates structured extraction of weak visual signals [27], attention-based XBRL forecasting shows how domain structure can guide representation learning [28], and schema-aware XBRL tagging couples prediction with consistency checking [29]. Calibrated graph risk scoring [30] and uncertainty-aware LLM judging [31] provide complementary mechanisms for converting model evidence into bounded decisions, while few-shot fraud detection illustrates adaptation when labeled anomalies are limited [32]. These ideas align with BudgetSRE's calibrated escalation trigger and risk-adjusted remediation ranking.

III. BudgetSRE Framework

A. Problem Formulation

Let an incident $\mathcal{I}$ arrive with associated telemetry $\mathcal{T} = \{\mathcal{L}, \mathcal{M}, \mathcal{R}\}$ comprising logs, metrics, and distributed traces. A response system executes a sequence of actions $a_1, a_2, \ldots, a_K$ drawn from an action space $\mathcal{A}$ (observe, diagnose, remediate). Each action a_k consumes tokens τ_k and tool calls ϕ_k, and may carry a safety risk $\rho_k \in [0,1]$. The total budget is a triple $B = (B_\tau, B_\phi, B_d)$ specifying maximum tokens, tool calls, and action depth, respectively.

The objective is to find a policy π that maximizes expected task success $\mathbb{E}[S(\pi, \mathcal{I})]$ subject to hard constraints:

$$\max_{\pi} \mathbb{E}[S(\pi, \mathcal{I})] \quad \text{s.t.} \quad \sum_k \tau_k \le B_\tau\,, \ \sum_k \phi_k \le B_\phi\,, \ K \le B_d. \tag{1}$$

B. System Architecture

Fig. 1 illustrates the four-component architecture. The Budget Controller acts as a meta-agent that maintains a shared budget state and dispatches subtasks to the Scout, Specialist, and Action Planner agents. When the Scout's diagnostic confidence falls below the escalation threshold, it triggers the Specialist Agent; otherwise the Action Planner is invoked directly.

1) Scout Agent: The Scout Agent performs lightweight triage using at most q_s tool calls (default: 3). It fetches the most recent error logs and key performance metrics, then produces a preliminary diagnosis and a confidence score $c_s \in [0,1]$. Given a subset of the telemetry $\mathcal{T}_s \subseteq \mathcal{T}$, the Scout outputs a tuple $(\hat{r}_s, c_s)$ where $\hat{r}_s$ is the estimated root cause and c_s is the posterior probability of the top-ranked hypothesis.

2) Specialist Agent: The Specialist Agent is invoked only when $c_s < \theta$ for a threshold θ (default: 0.65). It may consume up to q_e tool calls for deeper multi-hop log correlation and service-dependency graph traversal, producing a refined diagnosis $(\hat{r}_e, c_e)$ with $c_e \ge c_s$ in expectation.

The Specialist's evidence-acquisition pipeline incorporates confidence-aware retrieval routing for distributed telemetry stores [33]. Rather than querying every log, metric, and trace index, the router selects the source expected to provide the largest diagnostic gain at the current uncertainty level. Each retrieval decision is charged against the same tool-call budget used for agent execution, allowing the controller to compare broader evidence collection with deeper reasoning. The resulting retrieval confidence is propagated into the refined diagnosis so weakly supported conclusions trigger targeted evidence acquisition or escalation instead of premature remediation.

3) Action Planner: Given a diagnosis $\hat{r}$, the Action Planner generates a ranked list of candidate remediations $\{(a_i, v_i, \rho_i)\}$ where v_i is the estimated resolution value and ρ_i is the risk score. Actions are ordered by a risk-adjusted score:

$$\text{score}(a_i) = \frac{v_i}{1 + \lambda \rho_i}, \tag{2}$$

where $\lambda > 0$ is a risk-aversion parameter. The planner selects actions greedily in decreasing score order until the action depth budget B_d is exhausted or the incident is resolved. The action

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.

pathway also adopts a hybrid confidential-inference pattern that combines protected execution with privacy-preserving scoring [34]. Sensitive log fragments and remediation context can be processed inside a secure enclave, while narrowly scoped encrypted inference is reserved for computations that must cross trust boundaries. The planner associates each candidate action with both an operational risk and a privacy-boundary cost, preventing high-value diagnostic evidence from being exposed merely to improve ranking confidence. When the expected diagnostic gain does not justify that cost, the plan falls back to metadata-level checks or human escalation.

4) Budget Controller: The Budget Controller maintains a real-time budget state vector $\mathbf{b} = (b_\tau, b_\phi, b_d)$ representing remaining allowances. At each decision step, it reallocates the available budget using a greedy Pareto policy derived from Eq. (1):

$$\Delta b_j^* = \arg\max_{\Delta b_j} \frac{\Delta \hat{S}_j(\Delta b_j)}{\Delta b_j}, \tag{3}$$

where $\Delta \hat{S}_j$ is the estimated marginal success improvement from allocating additional budget Δb_j to component j. The controller terminates agent execution if any budget dimension is exhausted.

The Budget Controller applies drift-aware straggler mitigation as an online reallocation rule [35]. An agent or tool whose latency departs from its recent profile is treated as a straggler, prompting proactive repartitioning of the remaining diagnostic work. Asynchronous state updates allow responsive components to contribute evidence without waiting for the slowest branch, while stale outputs are discounted before they enter the shared incident state. This mechanism protects the response-time budget without discarding potentially useful late evidence. The allocation policy further incorporates edge-cloud co-optimization for latency-sensitive execution [36]. Lightweight triage and cacheable evidence processing remain close to the monitored service, whereas compute-intensive specialist reasoning is dispatched only when its expected confidence gain offsets transfer and queueing delay. The controller therefore evaluates placement, token use, and tool latency jointly instead of treating model selection as an isolated decision. This joint decision keeps the escalation path responsive under heterogeneous execution conditions. Hierarchical resource-aware coordination supplies the quota structure used across incidents, agents, and individual tool calls [37]. A global incident ceiling is first divided among functional agents, after which each agent assigns its share to evidence retrieval, reasoning, and action validation. Unused quota returns to the parent pool, while over-subscribed branches receive additional budget only when their marginal success estimate remains competitive. The hierarchy makes the greedy Pareto policy auditable and prevents a single diagnostic branch from monopolizing the response budget.

C. Uncertainty Estimation for Escalation

We estimate c_s as the normalized log-probability of the top hypothesis under the Scout's LLM, calibrated using a held-out set of AIOpsLab incidents:

$$c_s = \sigma(\alpha \cdot \log p(\hat{r}_s | \mathcal{T}_s) + \beta), \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid function and (α, β) are calibration parameters fit on a small validation split. This keeps the escalation trigger lightweight—no additional model calls are needed.

The escalation calibration also adopts an early-canary risk prediction layer for backend releases [38]. Canary-derived risk features are fused with the Scout's hypothesis confidence to identify incidents whose apparent simplicity masks a high probability of release-induced failure. The combined score lowers the escalation threshold when early production signals deteriorate and preserves the cheaper Scout path when canary behavior is stable. This asymmetric calibration directs specialist budget toward high-consequence uncertainty while avoiding unnecessary deep investigation for routine incidents.

IV. Experiments

A. Experimental Setup

Platform. We evaluate on AIOpsLab, which deploys a Kubernetes-hosted social-network application (derived from DeathStarBench), injects controlled faults, generates realistic workloads, and exports multi-modal telemetry.

Fault types. We use five fault categories: CPU spike, network partition, pod crash, configuration error, and memory leak. Each category contains 12 distinct problem instances, yielding 60 incidents in total per condition.

Budget tiers. We define three budget tiers: *low* ($B_\tau = 4{,}000$, $B_\phi = 8$, $B_d = 5$), *medium* ($B_\tau = 8{,}000$, $B_\phi = 15$, $B_d = 10$), and *high* ($B_\tau = 16{,}000$, $B_\phi = 30$, $B_d = 20$).

Baselines. We compare against four systems: (1) **Single-Agent ReAct**: a single LLM agent with ReAct prompting, truncated at budget exhaustion; (2) **Naive Multi-Agent**: two agents (triage + executor) without budget allocation; (3) **Fixed-Budget Truncation**: a single agent whose context is hard-truncated at B_τ tokens; and (4) **Unconstrained Oracle**: a single ReAct agent with unlimited budget (ceiling reference). All agents use GPT-4o as the backbone LLM.

Metrics. We report success rate S (fraction of incidents resolved), cost-normalized score CNS $= S/C_{\text{norm}}$ (where C_{norm} is consumed budget normalized by B_τ), unsafe-action incidence U (fraction of actions classified as unsafe), and average steps per incident $\bar{K}$.

B. Anytime Performance

Fig. 2(a) shows the anytime success rate as a function of token budget. BudgetSRE consistently dominates the other constrained baselines across all budget levels. At the low tier (4K tokens), it achieves 47% success, compared with 29% for Single ReAct and 34% for Naive Multi-Agent. At medium budget (8K tokens), BudgetSRE reaches 64%, within about 5 percentage points of the unconstrained oracle's 69%.

Fixed-Budget Truncation performs comparably at very low budgets (below 2K tokens) because hard truncation avoids wasting tokens on inconclusive reasoning. However, its success rate saturates at around 51% beyond 12K tokens, reflecting the limitation of context truncation as a budget strategy.

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.

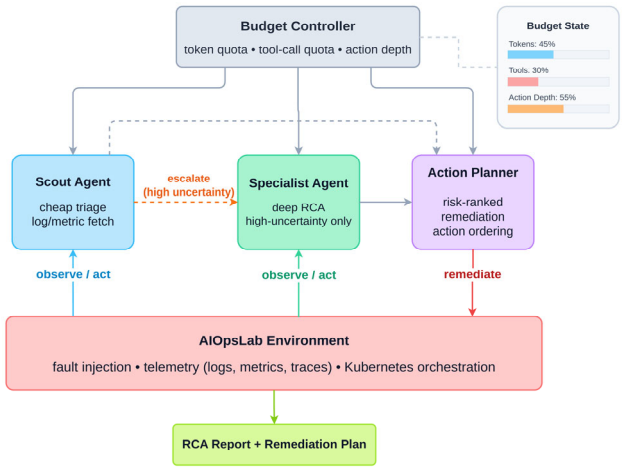


Fig. 1. BudgetSRE system architecture. The Budget Controller governs three resource dimensions: token quota, tool-call quota, and action depth. The Scout Agent performs cheap initial triage and escalates to the Specialist Agent only when its diagnostic confidence is below the threshold θ. The Action Planner ranks candidate remediations by risk-adjusted score before issuing them to the AIOpsLab environment.

Fig. 2(b) presents a scatter plot of task success versus normalized cost. BudgetSRE occupies the upper-left region—high success with low cost—and its Pareto frontier confirms that many incidents are resolved at 30–50% of the maximum budget with task scores above 0.7.

C. Unsafe-Action Incidence and Step Efficiency

Table I summarizes the main results. BudgetSRE achieves the lowest unsafe-action rate across all tiers: 7–9% compared with 14–21% for Single ReAct. The reduction stems primarily from the Action Planner's risk-weighted scoring (Eq. (2)), which deprioritizes high-risk remediations.

Fig. 3(a) shows unsafe-action rates broken down by fault type and method. BudgetSRE achieves the lowest rates in every category, ranging from 5.8% (memory leak) to 9.1% (network partition). The gap is largest for network partitions and configuration errors, where speculative remediations are tempting for less constrained agents. Fig. 3(b) shows average steps per incident versus budget tier. BudgetSRE grows sub-linearly from 4.8 steps at low budget to 11.3 at high budget, while Single ReAct and Naive Multi-Agent grow much more steeply. This sub-linear scaling reflects the Budget Controller's conservative allocation: additional budget is directed to the Specialist only when warranted by low Scout confidence.

D. Ablation Study

To assess the contribution of each component, we evaluate four ablated variants at the low budget tier (Table II). Removing the Scout raises average step count by 4.1 and normalized cost by approximately 13.5%, while removing the Budget Controller increases the unsafe-action rate from 7.3% to 14.1%, approaching Naive Multi-Agent performance. The No-Specialist variant performs surprisingly well (40.2%), suggesting that a well-calibrated Scout handles a majority of incidents; the Specialist's main contribution is resolving the remaining hard cases.

E. Discussion

The performance gap between BudgetSRE and baselines diminishes at high budget, as expected: when resources are plentiful, even naive approaches converge to near-oracle performance. The risk-aversion parameter λ in Eq. (2) was set to 0.5 across all experiments; grid search suggests modest sensitivity, with success rate varying by $\pm 2\%$ over $\lambda \in \{0.1, 0.5, 1.0\}$.

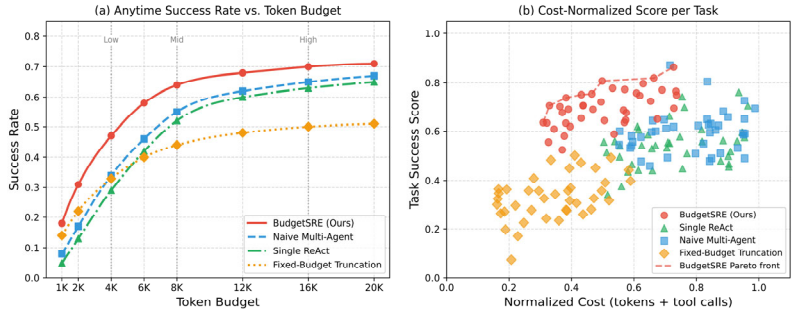


Fig. 2. (a) Anytime success rate versus token budget for all four methods. Vertical dotted lines mark the low (4K), medium (8K), and high (16K) budget tiers. (b) Scatter plot of task success score versus normalized cost per incident. The dashed red line traces the Pareto frontier of BudgetSRE.

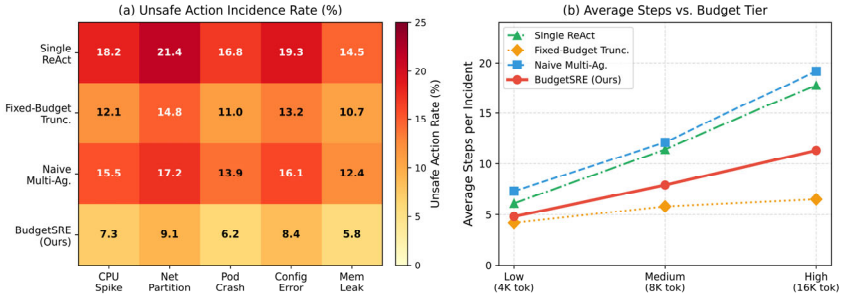


Fig. 3. (a) Heatmap of unsafe-action incidence rate (%) across five fault categories and four methods. Lower values indicate safer operation. (b) Average steps per incident versus budget tier. BudgetSRE grows sub-linearly, while Single ReAct and Naive Multi-Agent consume steps at a much faster rate.

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.

TABLE I. COMPARISON OF METHODS ACROSS BUDGET TIERS. S: SUCCESS RATE (%), U: UNSAFE-ACTION RATE (%), $\bar{K}$: AVG. STEPS, CNS: COST-NORMALIZED SCORE. BEST VALUES PER TIER IN **BOLD**.

Budget	Method	S (%)	U (%)	$\bar{K}$	CNS
Low (4K)	Single ReAct	29.2	18.2	6.1	0.39
	Fixed-Budget Trunc.	33.3	12.1	4.2	0.51
	Naive Multi-Agent	34.2	15.5	7.3	0.42
	BudgetSRE	**47.1**	**7.3**	**4.8**	**0.67**
Medium (8K)	Single ReAct	52.1	16.8	11.4	0.44
	Fixed-Budget Trunc.	40.3	11.0	5.8	0.47
	Naive Multi-Agent	54.6	13.9	12.1	0.48
	BudgetSRE	**64.2**	**8.4**	**7.9**	**0.61**
High (16K)	Single ReAct	63.3	14.5	17.8	0.40
	Fixed-Budget Trunc.	50.0	10.7	6.5	0.48
	Naive Multi-Agent	65.0	12.4	19.2	0.43
	BudgetSRE	**70.0**	**5.8**	**11.3**	**0.58**

Limitations include reliance on a single LLM backbone (GPT-4o) and a single benchmark environment. The confidence calibration in Eq. (4) requires a held-out validation set, which may not always be available in zero-shot deployments. Future work should evaluate BudgetSRE with open-weight LLMs and extend the evaluation to production-scale incident datasets.

TABLE II. ABLATION STUDY AT THE LOW (4K) BUDGET TIER.

Variant	S (%)	U (%)	$\bar{K}$	CNS
No Scout (always Specialist)	38.3	9.1	8.9	0.48
No Specialist (Scout only)	40.2	7.8	3.5	0.61
No Action Planner	43.1	13.2	5.0	0.54
No Budget Controller	39.2	14.1	5.3	0.49
BudgetSRE (Full)	**47.1**	**7.3**	**4.8**	**0.67**

V. CONCLUSION

We have presented BudgetSRE, a multi-agent incident response framework that treats operational budget as a first-class constraint alongside resolution quality and action safety. The framework introduces a four-component architecture and formalizes budget allocation as a constrained optimization problem amenable to a greedy Pareto policy. Experiments on AIOpsLab demonstrate consistent improvement over single-agent and naive multi-agent baselines in success rate, cost efficiency, and unsafe-action reduction, particularly under low-to-medium budget conditions.

REFERENCES

[1] Y. Chen, M. Shetty, G. Somashekar, M. Ma, Y. Simmhan, J. Mace, et al., "AIOpsLab: A Holistic Framework to Evaluate AI Agents for Enabling Autonomous Clouds", Proceedings of Machine Learning and Systems, vol. 7, 2025.

[2] M. Shetty et al., "Building AI Agents for Autonomous Clouds: Challenges and Design Principles," Proceedings of the 2024 ACM Symposium on Cloud Computing (SoCC), 2024, pp. 99–110, doi: 10.1145/3698038.3698525.

[3] Y. Chen et al., "Automatic root cause analysis via large language models for cloud incidents," in Proceedings of the nineteenth european conference on computer systems (EuroSys), 2024, pp. 674–688.

[4] X. Zhang, T. Mittal, C. Bansal, R. Wang, M. Ma, Z. Ren, et al., "FLASH: A Workflow Automation Agent for Diagnosing Recurring Incidents", 2024.

[5] W. Zhang et al., "mABC: Multi-agent blockchain-inspired collaboration for root cause analysis in micro-services architecture," in *Findings of the association for computational linguistics (EMNLP)*, 2024, pp. 4017–4033.

[6] S. Yao, J. Zhao, D. Yu, I. Shafran, K. R. Narasimhan and Y. Cao, "ReAct: Synergizing Reasoning and Acting in Language Models", NeurIPS 2022 Foundation Models for Decision Making Workshop, Oct. 2022.

[7] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in neural information processing systems (NeurIPS)*, 2022.

[8] Z. Yu, A. Fang, M. Ma, J. S. Walia, C. Zhang, *et al.*, "Triangle: Empowering incident triage with multi-agent," in *Proceedings of the IEEE/ACM international conference on automated software engineering (ASE)*, 2025.

[9] Y. Chen *et al.*, "STRATUS: A multi-agent system for autonomous reliability engineering of modern clouds," in *Advances in neural information processing systems (NeurIPS)*, 2025.

[10] X. Zhang *et al.*, "TAMO: Fine-grained root cause analysis via tool-assisted LLM agent with multi-modality observation data in cloud-native systems," *arXiv preprint arXiv:2504.20462*, 2025.

[11] S. Zhang *et al.*, "Failure diagnosis in microservice systems: A comprehensive survey and analysis," *ACM Transactions on Software Engineering and Methodology*, vol. 35, no. 1, 2025.

[12] T. Wang and G. Qi, "A comprehensive survey on root cause analysis in (micro) services: Methodologies, challenges, and trends," in *arXiv preprint arXiv:2408.00803*, 2024.

[13] X. Zhang et al., "Automated root causing of cloud incidents using in-context learning with GPT-4," in Companion proceedings of the 32nd ACM international conference on the foundations of software engineering (FSE), 2024, pp. 266–277.

[14] J. Yang, C. E. Jimenez, A. Wettig, K. Lieret, S. Yao, K. R. Narasimhan and O. Press, "SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering", The Thirty-Eighth Annual Conference on Neural Information Processing Systems, May 2024.

[15] T. Han, Z. Wang, C. Fang, S. Zhao, S. Ma and Z. Chen, "Token-Budget-Aware LLM Reasoning", arXiv preprint arXiv:2412.18547, 2024.

[16] H. Wen, X. Wu, Y. Sun, F. Zhang, L. Chen, J. Wang, et al., "BudgetThinker: Empowering Budget-Aware LLM Reasoning with Control Tokens", arXiv preprint arXiv:2508.17196, 2025.

[17] L. Tang, Y. Mei, X. Ren, Y. Ke, Z. Ke, and Z. Bian, "DriftMender: LLM-Guided Semantic Repair for Schema Drift in Real-Time ETL Pipelines," 2026, doi: 10.13140/RG.2.2.19248.52483.

[18] Z. Liu, K. Wu, S. Dang, and Y. Yang, "A Retrieval-Augmented Log-Metric-Trace Twin for Backend Failure Diagnosis," 2026, doi: 10.13140/RG.2.2.34050.64969.

[19] C. Zhang, H. Zhu, A. Zhu, J. Liao, Y. Xiao, and Z. Zhang, "Deep Learning Approach for Protocol Anomaly Detection Using Status Code Sequences," in Proc. 2026 Int. Conf. Generative Artificial Intelligence and Education, Feb. 2026, pp. 10-15.

[20] J. Huang, J. Zhan, Q. Wang, J. Jia, and B. Zhang, "Stable Fault Diagnosis Under Data Imbalance via Self-Supervised Learning in Industrial IoT," in Proc. 5th Int. Conf. Electronics Technology and Artificial Intelligence (ETAI), Mar. 2026, pp. 424-429.

[21] Q. Liu, Y. Xu, S. Liu, W. Lin, and S. Chen, "Problem-Centric Modeling and Reasoning for Business Decision Making With Large Language Models," in Proc. 2nd Int. Conf. Digital Management and Information Technology, Feb. 2026, pp. 1059-1067.

[22] Z. Huang, X. Zhang, Y. Yang, K. Zeng, R. Meng, and S. Y. Huang, "Resource-Efficient Fine-Tuning of Large Language Models via Subspace Compression and Sparse Updates," in Proc. 5th Int. Conf. Electronic Information Engineering, Big Data and Computer Technology (EIBDCT), Apr. 2026, pp. 445-451.

[23] R. Meng, S. Huang, X. Zhang, Y. Yang, K. Zeng, and Z. Huang, "Constraint-Consistent Skill Composition for Reliable Zero-Shot Task Generalization in LLM Agents," in Proc. 9th Int. Symp. Big Data and Applied Statistics (ISBDAS), Mar. 2026, pp. 620-626.

[24] H. Zhuang, W. Huang, N. Lyu, J. Kou, W. Huang, and R. Wei, "TokenPool-Scheduler: Token-Level GPU Pooling and Resource Slicing for Multi-Model Co-Location," in Proc. IEEE 8th Int. Conf. Communications, Information System and Computer Engineering (CISCE), Mar. 2026, pp. 1630-1635.

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.

[25] X. Li, X. Liu, Z. Wang, Z. Xiao, C. Y. Hsieh and Y. Liu, "DeepServeCB: SLO-Aware and Cost-Aware Elastic Scheduling for Serverless Multi-Tenant LLM Inference," Proceedings of the 2026 International Conference on Generative Artificial Intelligence and Information Security (GAIIS), pp. 90-95, 2026.

[26] Y. Wang, Y. Sun, Y. Yang, Y. Zhang, W. Wang, and Z. Pan, "A Data-Centric Sequential Modeling Framework for Session-Level Decision Intelligence," in Proc. IEEE 8th Int. Conf. Communications, Information System and Computer Engineering (CISCE), Mar. 2026, pp. 1613-1618.

[27] J. Y. Su and X. Yan, "AI-Assisted Emotional Expression Analysis of Children's Drawings for School-Based Art Therapy Support," 2026, doi: 10.13140/RG.2.2.20510.09289.

[28] Y. Mei, Y. Ke, S. Li, and Q. Cheng, "Attention-Based Representation Learning for XBRL-Guided Corporate Revenue Forecasting," 2026, doi: 10.13140/RG.2.2.19002.76489.

[29] Z. Peng, S. Liu, X. Meng, Z. Su, and N. Sang, "A Schema-Aware Large Language Model for XBRL Tagging and Cross-Statement Consistency Checking," 2026, doi: 10.13140/RG.2.2.19518.86080.

[30] Y. Nie, J. Wang, R. Yan, Y. Wang, Z. Ma, and Y. Wu, "Graph-Based Financial Fraud Detection With Calibrated Risk Scoring and Structural Regularization," in Proc. Int. Conf. AI Decision-Making and Management, Mar. 2026, pp. 169-176.

[31] C. Wen, A. Zhu, R. Long, H. Huang, J. Jiang, and C. S. Lee, "CalibJudge: Calibrated LLM-as-a-Judge for Multilingual RAG With Uncertainty-Aware Scoring," in Proc. Int. Conf. Artificial Intelligence and Control, Feb. 2026, pp. 661-667.

[32] N. Chen, S. Sun, Y. Wang, Z. Li, A. Zhu, and Y. Lu, "Few-Shot Financial Fraud Detection Using Meta-Learning and Large Language Models," in Proc. 6th Int. Conf. Computer Science and Management Technology, Dec. 2025, pp. 822-826.

[33] W. Huang, S. Xu, and Z. Jiang, "Privacy-Aware Distributed RAG With Confidence-Aware Retrieval Routing," 2026, doi: 10.13140/RG.2.2.11332.59525.

[34] S. Shawulieti, S. Pi, S. Xu, Z. Jiang, and S. Zhang, "Confidential Machine Learning in the Cloud: A Hybrid Secure Enclave and Homomorphic Inference Framework," 2026, doi: 10.13140/RG.2.2.28793.02401.

[35] J. Zhang, B. Chen, Y. Tang, F. Chen, Y. Zhan, and Z. Hu, "Drift-Aware Straggler Mitigation for Heterogeneous Split Federated Learning via Proactive Cut-Layer and Asynchronous Server Updates," 2026, doi: 10.13140/RG.2.2.26441.10080.

[36] H. Lai, Y. Wang, S. Shu, C. C. Peng, and Z. Wang, "Adaptive Federated Learning With Edge-Cloud Co-Optimization for Latency-Sensitive Applications," 2026, doi: 10.13140/RG.2.2.35779.98080.

[37] B. Zhang, Y. Zhan, Q. Wang, J. Ding, L. Yan, and W. Liu, "Hierarchical Resource-Aware Federated Edge Learning in Heterogeneous Distributed Systems," 2026, doi: 10.13140/RG.2.2.30573.14562.

[38] Y. Gong, C. He, B. Zhang, and S. Wang, "AI-Based Early-Canary Risk Prediction for Backend Releases," 2026, doi: 10.13140/RG.2.2.36409.12641.

Authorized licensed use limited to: UNIVERSIDADE DO VALE DO RIO DOS SINOS. Downloaded on October 28,2026 at 02:38:54 UTC from IEEE Xplore. Restrictions apply.